

# JOHN CARROLL

## Forward Deployed Engineering | Enterprise Platforms | Agentic Systems

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AI architecture and engineering leader in The Cigna Group / Evernorth's AI Enablement Office, leading enterprise initiatives across agentic systems, retrieval, platform delivery, and business-aligned AI solutions.

### PROFESSIONAL EXPERIENCE

#### Cigna | Evernorth (via Solomon Page) - Senior Principal Architect

Nov 2025 - Present | Atlanta, GA

- Lead applied AI architecture and forward deployed engineering for the AI Enablement Office, embedding with business units to identify operational problems, build AI prototypes, and convert high-value POCs into scalable enterprise platforms.
- Help establish the internal Forward Deployed Engineering model, including the FDE playbook, hiring guidance, delivery structure, and repeatable operating model for business-aligned AI solution delivery.
- Architect enterprise AI systems across agentic workflows, RAG, search/retrieval, analytics integrations, evaluation, and governance, translating business needs into secure, reliable, and production-ready technical designs.
- Guide engineering execution for an FDE pod by coordinating developers, setting technical direction, defining phases, managing roadmap tradeoffs, and aligning implementation commitments with stakeholder expectations.
- Design and implement retrieval pipelines using ingestion, chunking, indexing, grounding, semantic/hybrid search, evaluation, and quality monitoring to improve answer reliability, traceability, and enterprise reuse.
- Build production services and full-stack prototypes integrating LLM tool use, safety guardrails, measurable evaluation, ETL pipelines, knowledge graph-assisted retrieval, multimodal workflows, and translation capabilities.
- Support internal AI platform strategy and build-vs-buy decisions by evaluating enterprise AI patterns, reusable architecture, data governance needs, and scalable internal delivery models.
- Enable executive-facing AI adoption through senior leadership demos, stakeholder education, and delivery leadership for production AI systems.

#### Microsoft - Senior Solution Engineer | AI Apps & Agents Global Black Belt

Sep 2023 - Nov 2025 | Atlanta, GA

- Trusted advisor and architect for Fortune 500 clients, driving \$200M+ in AI-related revenue within my first year by shaping AI adoption strategies, leading technical deep dives, and unblocking enterprise-scale deployments for companies including Nvidia, Walmart, Coke, and Cisco.
- Engineer and deploy production-grade agentic AI systems that plan, reason, and take safe, reliable action across mission-critical workflows—balancing business requirements with technical execution.

### EDUCATION

#### Georgia Institute of Technology

M.S. Computer Science

AI Specialization

Expected Start: August 2026

#### Quantic School of Business and Technology

Executive MBA

March 2025 - Expected July 2026

#### Auburn University

B.S. Software Engineering

August 2011 - May 2016

### CERTIFICATIONS

Neo4j Graph Data Science (Nov 2024)

Azure AI Engineer Associate (Dec 2025)

### RECOGNITION

#### Lead Engineer - Cigna

Fastest Production Release

January 2026

#### Judge - Microsoft

Global Hackathon, AI Agents Category

August 2025

#### Lead Developer - Microsoft

AI Global Black Belt Hackathon

1st Place, April 2024

#### Team Lead - Boeing

Alabama 20th Annual STARS Award

"Predictability and Stability"

February 2023

### SKILLS

**Core:** Applied AI Engineering; Full-Stack Development; DevSecOps; AI Product

- Designed and shipped retrieval-augmented generation (RAG) pipelines, including indexing and chunking strategies, semantic search, and grounding, which improved precision and recall for enterprise search and automation.
- Built AI agents using LangChain, LangGraph, Semantic Kernel, AutoGen, and PromptFlow, implementing tool-use orchestration, memory, safety guardrails, and cost/latency tradeoff strategies to ensure enterprise-grade reliability.
- Leveraged evaluation frameworks and AI orchestration at the network level (Azure API Management / API Center AI Gateway) to optimize performance, safety, and efficient model utilization across high-scale deployments.
- Delivered end-to-end AI solutions (computer vision, OCR, NLP-to-SQL, video indexing) that automated decision-making, reduced operational costs, and accelerated digital transformation initiatives.
- Created reusable accelerators, POCs, and solution IP adopted by clients and Microsoft field teams, multiplying impact and standardizing patterns for scalable customer-facing AI deployments.

### Boeing - Senior AI/ML Software Engineer

*Jun 2019 - Sep 2023 | Huntsville, AL*

- Spearheaded value stream data visualization and analytics solution focused on design and procurement processes, enhancing business intelligence for high-profile programs (EUS, SLS, ISS, PAC3, GMD). Delivered \$25M in annual efficiency savings, earning Alabama's 20th STARS Award.
- Headed the redesign of a 3D factory layout Unity app, optimizing UX and boosting operator efficiency, resulting in a 25% increase in conversion rates and a 15% reduction in user churn.
- Implemented a root cause analysis solution for part changes in engineering processes, driving key insights through metrics, substantially reducing subsystem errors for 1,000+ daily users.
- Engineered a Natural Language Processing (NLP)-based authoring tool for aircraft requirements, enhancing accuracy, efficiency, and reducing ambiguity, as part of an initiative aimed at achieving \$10B in annual cost savings across the enterprise by minimizing revision cycles.

### Hexagon Safety & Infrastructure - Full-Stack Software Engineer

*Jun 2016 - Jul 2019 | Madison, AL*

- Engineered Smart Cities public safety SaaS on Azure, integrating IoT, robot/drone command-and-control, and computer vision-powered video analytics across real-time, scalable, multi-tenant architecture.
- Led DevSecOps execution for securing cloud resources and delivery pipelines; built and maintained CI/CD automation and infrastructure-as-code, improving reliability and deployment consistency.
- Supported compliance/quality standards (e.g., Microsoft Gold Partner expectations, ISO 9001 practices) through robust engineering processes and operational rigor.

Prototyping → Production; Data Analytics & Visualization; Agile Delivery (Scrum/Kanban)

**Languages:** Python; TypeScript/ JavaScript; C#; SQL; Node.js; Bash; Java

**Applied AI Systems:** LLM applications; RAG; agentic orchestration; tool/ function calling; prompt design; safety guardrails; evaluation/ benchmarks; quality monitoring; human-in-the-loop workflows

**Retrieval & Search:** chunking/indexing strategies; hybrid search; reranking; grounding/citations; query rewriting; caching; latency/cost tradeoffs

**Data & Knowledge:** knowledge graphs; entity linking; schema/ontology modeling; embeddings; vector databases; metadata pipelines; ETL/ELT; data quality checks

**Backend & APIs:** REST/JSON APIs; authn/authz patterns; service integration; background jobs/queues, testing strategies; performance tuning

**Frontend:** React; Next.js; Angular; Vue; component-driven UI; accessibility basics; user-centered design

**Cloud/DevOps:** Azure; AWS; GCP; Docker; Kubernetes; CI/CD; infrastructure-as-code; secrets management; observability

**Frameworks/Tools:** LangChain; LangGraph; LangSmith; Grafana; AutoGen; GitHub; Linear; Cursor, Claude Code; Codex

## **ADDITIONAL EXPERIENCE**

**Auburn University - Computer Science Teaching Assistant** *Aug 2015 - May 2016 | Auburn, AL*

**Neptune Technology Group - Software Engineer (Co-op Program)** *Jan 2013 - May 2015 | Tallassee, AL*

*(Completed three software/electrical engineering co-op rotations.)*

**Independent Consultant & Full-Stack Software Engineer** *Jan 2011 - Present*